Analyzing the Impact of Wi-Fi Usage on Student Productivity Using Data Analytics

Panuganti Vishishta
panugativish@iitbhilai.ac.in
Indian Institute of Technology Bhilai
Durg, Chhattisgarh, India

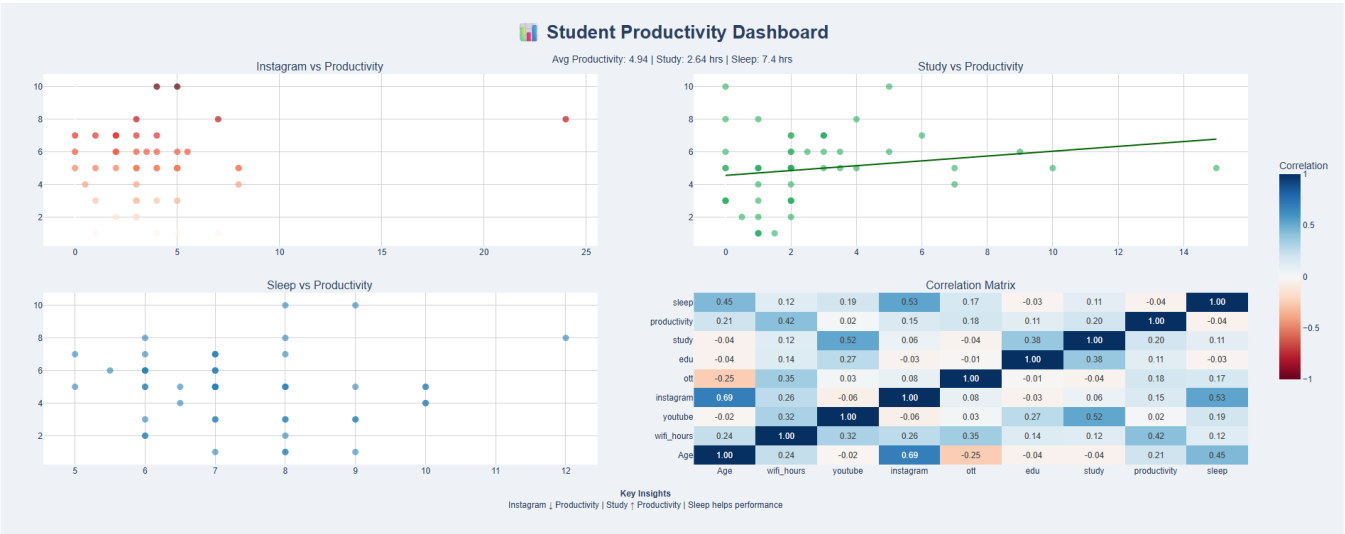

Figure 1: Student Productivity Dashboard showing relationships between study habits, social media usage, and productivity

Abstract

This study explores how Wi-Fi usage influences student productivity, study habits, and overall behavior. With the increasing reliance on internet connectivity, students benefit from easy access to educational resources but are also exposed to distractions such as social media and entertainment platforms.

The dataset used in this study was collected through survey responses and includes factors such as Wi-Fi usage hours, social media usage, study time, sleep patterns, and self-reported productivity levels. Data preprocessing techniques were applied to handle inconsistencies and convert raw inputs into a structured format suitable for analysis.

Exploratory Data Analysis (EDA) was performed to identify patterns and relationships among variables. The analysis revealed that **higher social media usage is associated with lower productivity**, while **increased study time positively impacts performance**. A Linear Regression model was further developed to predict productivity based on these behavioral factors.

The findings suggest that although Wi-Fi is an essential tool for learning, **excessive and unregulated usage negatively affects student productivity**. Maintaining a balanced and disciplined approach to internet usage is therefore crucial for improving academic outcomes.

1 Introduction

1.1 Background

The widespread availability of Wi-Fi has significantly transformed the educational landscape by enabling access to digital learning resources, online courses, and collaborative tools. Students increasingly rely on internet connectivity for academic purposes; however, this accessibility also exposes them to various distractions such as social media platforms, streaming services, and online gaming. As a result, **understanding the impact of Wi-Fi usage on student productivity** has become an important area of study.

1.2 Problem Statement

Despite the advantages of internet access, students often struggle to maintain a **balance between productive and non-productive usage**. Excessive engagement in non-academic activities can lead to **reduced focus, lower efficiency, and decreased academic performance**. This study aims to analyze and quantify the relationship between internet usage patterns and student productivity, while also examining the influence of factors such as **study habits and sleep patterns**. Additionally, it seeks to identify the**key factors that contribute to distraction** among students.

2 Objectives

The main objectives of this study are as follows:

- To **collect and preprocess primary data** using a survey-based approach
- To gather **secondary insights through web scraping** of relevant articles
- To analyze student behavioral patterns using **Exploratory Data Analysis (EDA)**
- To **visualize relationships and trends** among key variables
- To develop a **predictive model** to estimate student productivity

3 Dataset Description

The dataset used in this study consists of survey responses collected from students to understand their **Wi-Fi usage patterns and their impact on productivity**. The data captures various aspects of student behavior, including demographic details, internet usage, academic habits, and lifestyle factors.

The dataset includes the following features:

- **Demographic Features:** Age and gender of the respondents.
- **Usage Metrics:** Daily Wi-Fi usage hours and time spent on platforms such as YouTube, Instagram, and OTT services.
- **Academic Metrics:** Study hours and time spent on **educational activities**.
- **Behavioral Metrics:** Sleep duration, common distraction factors, and **self-reported productivity levels** measured on a scale of 1 to 10.

These features provide a **comprehensive view of student behavior**, enabling analysis of how different usage patterns influence productivity.

4 Data Cleaning and Preparation

Real-world datasets often contain **inconsistencies, missing values, and irregular formats** that must be addressed before analysis. Therefore, several preprocessing steps were applied to ensure **data quality and reliability**.

4.1 Handling Missing Values

Missing numerical values were handled using **median imputation** to minimize the impact of extreme values. Additionally, non-numeric textual entries such as "depends" and "varies" were identified and converted into missing values before further processing.

4.2 Data Transformation

The dataset contained multiple inconsistencies in input formats. These were standardized into numerical values for analysis. For example, entries such as "5 hours" and "8hrs" were converted into numeric values, ranges like "6–8" were replaced with their **average (7)**, and expressions such as "Minimum 6" were interpreted as 6.

4.3 Outlier Detection and Removal

Outliers and unrealistic values were identified and removed to improve **data accuracy**. For instance, Wi-Fi usage exceeding 24 hours, sleep durations below 3 hours or above 12 hours, and extreme values such as 99 or negative entries were excluded from the dataset.

4.4 Data Normalization

Column names were standardized for consistency, and categorical variables such as gender and binary responses (Yes/No) were converted into appropriate **encoded formats** to facilitate analysis.

4.5 Duplicate Removal

Duplicate records were removed to avoid redundancy and ensure **data integrity**. This was implemented using the following operation:

```
df.drop_duplicates(inplace=True)
```

5 Exploratory Data Analysis and Visualization

To understand the factors affecting student productivity, a focused set of visualizations was selected. Each plot highlights a key relationship relevant to productivity.

5.1 Study Hours vs Productivity

The figure below shows a **clear positive relationship** between study hours and productivity. Students who spend more time studying tend to have higher productivity levels, indicating that **study habits play a significant role** in academic performance.

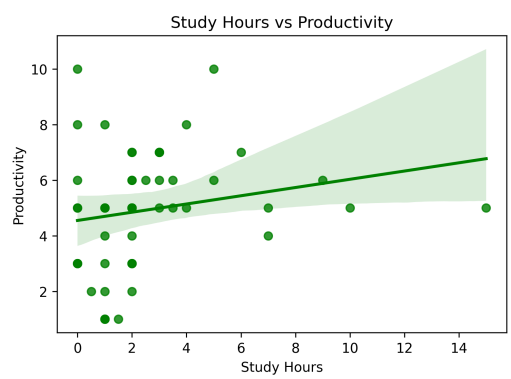

Figure 2: Relationship between Study Hours and Productivity

5.2 Instagram Usage vs Productivity

This plot demonstrates a **negative relationship** between Instagram usage and productivity. As time spent on Instagram increases, productivity tends to decrease, suggesting that **excessive social media usage acts as a major source of distraction**.

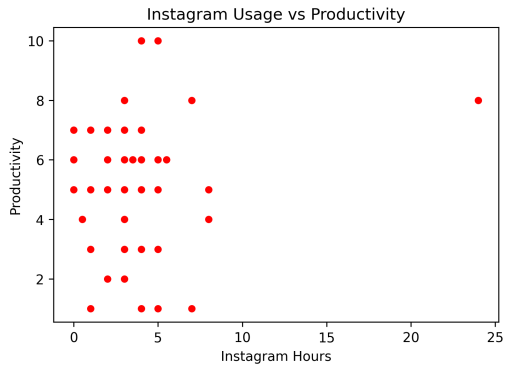

Figure 3: Impact of Instagram Usage on Productivity

5.3 Correlation Heatmap

The correlation heatmap provides a comprehensive view of relationships between variables. Study hours show a **strong positive correlation** with productivity, while Instagram usage has a **negative correlation**. Sleep duration exhibits a **moderate positive influence**.

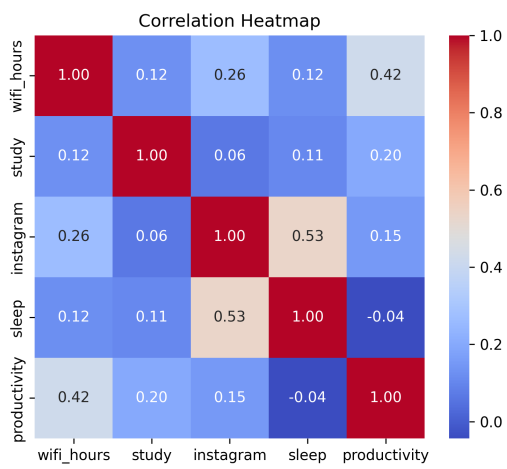

Figure 4: Correlation Matrix of Key Variables

5.4 Distribution of Productivity

The distribution plot indicates that most students fall within a **moderate productivity range**, with fewer students at the extreme ends. This suggests that productivity levels are generally balanced across the dataset.

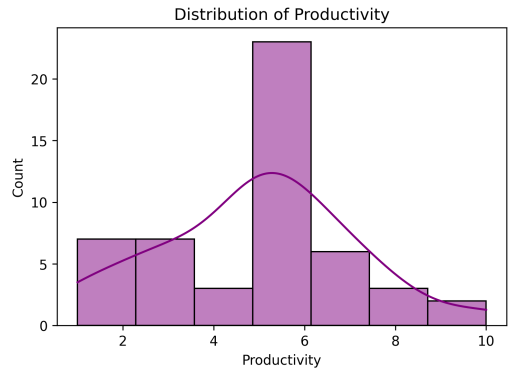

Figure 5: Distribution of Productivity Scores

5.5 Outlier Detection using Boxplot

The boxplot highlights the presence of **outliers in Instagram usage**. A small group of students spends significantly more time on social media compared to others, which may negatively impact their productivity.

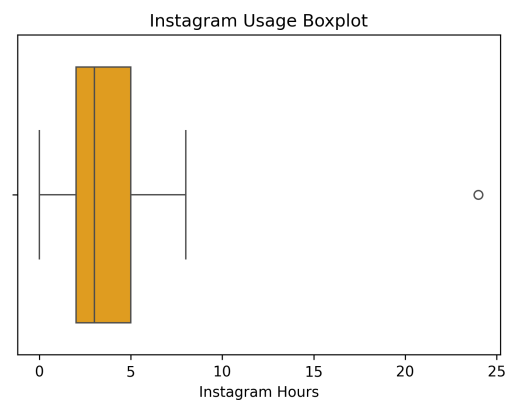

Figure 6: Boxplot of Instagram Usage

5.6 Summary of Insights

The selected visualizations clearly indicate that student productivity is influenced by **behavioral factors**. Study hours have a **strong positive effect**, while excessive social media usage negatively impacts performance. Sleep contributes moderately, emphasizing the importance of maintaining a **balanced routine**.

6 Dashboard Visualization

An interactive dashboard was developed using Plotly to provide a **comprehensive and intuitive view** of the relationship between student behavior and productivity. The dashboard combines multiple visualizations, including scatter plots and a correlation heatmap, to present **key insights in a structured manner**.

The visualizations clearly indicate that **study hours have a positive impact on productivity**, whereas **increased Instagram usage negatively affects performance**. Sleep duration shows a

moderate positive influence, highlighting its supporting role in maintaining consistent productivity levels.

Color-coded representations are used to improve interpretability: red indicates **negative impact**, green represents **positive influence**, and blue highlights **supporting factors**. Overall, the dashboard effectively summarizes the analysis and provides a **clear understanding of behavioral influences on productivity**.

7 Model Building

7.1 Objective

The objective of the model is to **predict student productivity** based on behavioral factors such as study habits, social media usage, sleep patterns, and Wi-Fi usage.

7.2 Model Selection

A **Linear Regression model** was selected for this study due to its simplicity, interpretability, and suitability for predicting continuous values such as productivity scores.

7.3 Feature Selection

The following features were used as input variables for the model:

- Study hours
- Instagram usage
- Sleep duration
- Wi-Fi usage

7.4 Model Implementation

The dataset was divided into **training and testing sets**. A Linear Regression model was trained using the selected features, and predictions were generated for the test data.

7.5 Model Evaluation

The model was evaluated using **Mean Squared Error (MSE)** and the coefficient of determination (R^2 **score**). The obtained R^2 score was **negative**, indicating that the model does not fit the data well and performs worse than a baseline mean prediction.

7.6 Interpretation

The model coefficients provide insight into the influence of each variable on productivity. **Study hours show a positive relationship** with productivity, indicating that increased study time improves performance. **Instagram usage has a negative impact**, suggesting that excessive social media usage reduces productivity. Sleep shows a **moderate influence**, while Wi-Fi usage contributes to both academic and non-academic activities.

Although the model captures general trends, its **low performance indicates missing influencing factors**, such as individual motivation, environment, and external conditions, which are not included in the dataset.

8 Project Repository

The complete project code, dataset, and implementation details are available at the following GitHub repository:

https://github.com/VishishtaPanuganti/Student-productivity-analysis

9 Conclusion

This study analyzed the impact of Wi-Fi usage and related behavioral factors on student productivity using survey data and analytical techniques. The results show that productivity is significantly influenced by **how students utilize their time**, rather than internet access alone.

Exploratory Data Analysis revealed that **study hours have a strong positive effect on productivity**, while **excessive Instagram usage negatively impacts performance**. Sleep was found to play a **supportive role** in maintaining consistent productivity levels.

The dashboard provided a **clear and intuitive visualization** of these relationships, enabling better understanding of the factors affecting student behavior. Although a Linear Regression model was developed, its **limited performance** indicates that productivity is influenced by additional factors beyond the selected variables.

Overall, the study highlights the importance of **balanced and disciplined Wi-Fi usage**. Students who effectively manage their study time and limit distractions tend to achieve better productivity outcomes. Future work can include additional variables such as academic environment, mental health, and motivation to improve predictive accuracy.